

FRA Diagnostic Analysis Report

Standard: IEEE C57.149-2012 / IEC 60076-18 Standards-Aligned SFRA Diagnostic Pipeline | **System:** AI-Powered Hybrid Transformer Integrity Monitor

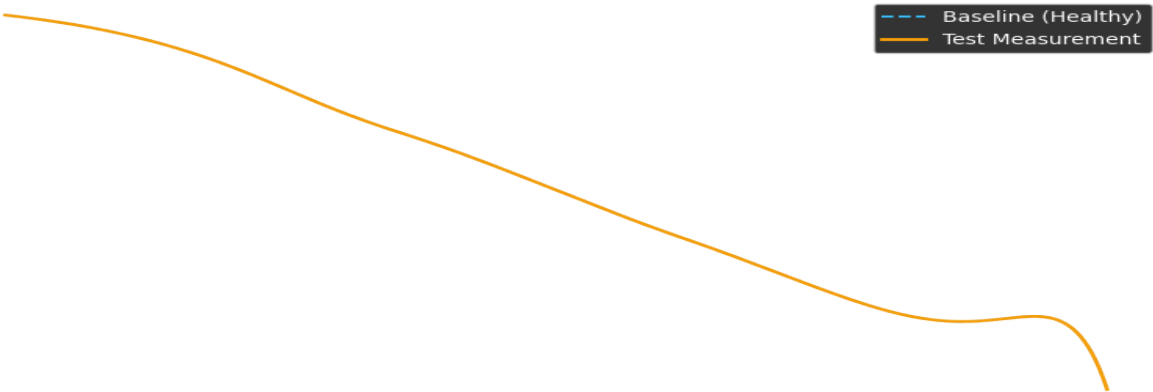
Asset / Transformer ID: TX-ASSET-01	Inspection Date: 2026-10-02 19:03
Winding Configuration: HV to LV Transfer Function	Overall Status / Severity: Warning (Medium)

Overall Health Integrity Score Blended Physics (60%) + ML (40%)	72.8%	Condition Assessment: Moderate Degradation Detected AI Fault Signature: Insulation Degradation (32.0% conf.) Overall Correlation (CCF): 1.0000 Max Dev: 0.00 dB
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Sub-band Diagnostic Breakdown (IEEE C57.149 Standards Table)

Sub-Band	Frequency Range	Correlation (CCF)	Max Deviation (dB)	Health Status
Low (Core)	< 2 kHz	1.0000	0.00 dB	Healthy
Mid (Winding)	2 - 100 kHz	1.0000	0.00 dB	Healthy
High (Insulation)	> 100 kHz	N/A	N/A	Insufficient Data

Frequency Response Bode Plot Comparison



Actionable Recommendations & Maintenance Guidance

Primary Assessment: Minor deviation detected. Schedule a DGA (Dissolved Gas Analysis) to confirm internal state.
Targeted Protocol: Moderate deviations observed in frequency response signature. Schedule oil sampling for Dissolved Gas Analysis (DGA) within 30 days. Re-sweep SFRA at next scheduled maintenance outage to monitor fault progression.
Component-Level Guidance (IEEE C57.149): Core (Low Band, < 2 kHz): Operating within IEEE C57.149 normal parameters. No physical deformation indicated. Winding (Mid Band, 2–100 kHz): Operating within IEEE C57.149 normal parameters. No physical deformation indicated. Insulation (High Band, > 100 kHz): Insufficient data points (< 10 points) in this sub-band. Re-sweep frequency response.

Deterministic Physics Floor: As mandated by IEEE C57.149-2012 and IEC 60076-18, the diagnostic status is governed strictly by the worst-performing sub-band (Low: Core, Mid: Winding, High: Insulation). Machine Learning predictions provide fault signature classification and escalation without overriding physical limits.